

A 0.4395 Upper Bound for the i.i.d. Berry–Esseen Constant

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Abstract

We prove that the absolute constant in the classical Berry–Esseen inequality for identically distributed summands is at most $879/2000 = 0.4395$, uniformly over all sample sizes and all centered variance-one laws with finite absolute third moment. The proof couples the symmetrized third-moment ratio $r = \mathbb{E}|X - X'|^3 / (2\mathbb{E}|X|^3)$ with $d = 1 - \mathbb{E}|X|$, where X' is an independent copy of X . Their joint constraints give a characteristic-function comparison with the symmetric two-point law and a stronger modulus estimate when d is small. Prawitz smoothing reduces the remaining argument to continuous-parameter inequalities, proved by exact outward-rounded dyadic certificates. The analytic reduction and the exact certificates are formally verified in Lean. The proof was developed autonomously using OpenAI Codex, primarily with GPT-5.6 Sol.

1 Introduction

Let $X, X_1, X_2, \dots$ be independent and identically distributed real random variables with

$$\mathbb{E}X = 0, \quad \mathbb{E}X^2 = 1, \quad \rho := \mathbb{E}|X|^3 < \infty,$$

and call a law satisfying these conditions admissible. Set

$$S_n := \frac{X_1 + \dots + X_n}{\sqrt{n}}, \quad \Delta_n := \sup_{x \in \mathbb{R}} |\mathbb{P}\{S_n \leq x\} - \Phi(x)|,$$

where Φ denotes the standard normal distribution function. The optimal i.i.d. Berry–Esseen constant is

$$C_0 := \inf \left\{ C > 0 : \Delta_n \leq C \frac{\rho}{\sqrt{n}} \text{ for all admissible laws and all } n \geq 1 \right\}.$$

Esseen’s classical example gives

$$C_0 \geq C_E := \frac{\sqrt{10} + 3}{6\sqrt{2\pi}} = 0.409732\dots [3].$$

Shevtsova proved the published upper bound $C_0 \leq 0.4690$ [7]; Bobkov’s survey records the resulting interval $0.4097 < C_0 < 0.4690$ [1, Theorem 20.1].

Theorem 1. *For every admissible law and every integer $n \geq 1$,*

$$\Delta_n \leq \frac{879}{2000} \frac{\rho}{\sqrt{n}}.$$

Consequently, $C_0 \leq 0.4395$.

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The starting point is the shared moment parameter

$$r := \frac{\mathbb{E}|X - X'|^3}{2\rho},$$

where X' is an independent copy of X . Symmetrization yields a modulus bound in terms of r , while a stop-loss identity gives a bound for the sine remainder $\mathbb{E}[\sin(uX) - uX]$ in terms of the same r . Since $\mathbb{E}X = 0$, this remainder is the imaginary part of the characteristic function. The resulting disk comparison yields the 0.45 bound in the earlier version of this note.

The present improvement retains more of the moment information. If $\eta = \rho(r - 1)$ and $d = 1 - \mathbb{E}|X|$, then

$$0 \leq d \leq \min(1 - \rho^{-1}, 1 - \eta).$$

When these bounds force d to be small, expansion about $|X| = 1$ controls the real and imaginary characteristic-function components separately. This adds a comparison with $\cos u$, the characteristic function of the symmetric two-point law, without assuming that X is symmetric. In the large-sample analysis it also gives an explicit exponential modulus rate. Keeping the admissible range of the Gaussian correction and the sample-size dependence in the telescoping bound further sharpens the comparison. Write $L = \rho/\sqrt{n}$.

A distribution-free bound handles $L \geq 1090/879$. Exact interval certificates cover the remaining finite- and large-sample regimes, including every boundary and arbitrarily small positive L .

2 Moment constraints for the characteristic function

The symmetrized third moment controls both the modulus and the imaginary part of the characteristic function. Retaining the first absolute moment gives an additional comparison with the symmetric two-point law.

2.1 Stop-loss identities

The stop-loss representation of the symmetrized third moment restricts r and relates it to the signed third moment, providing the real and imaginary constraints used below.

Let X' be an independent copy of X , and define

$$m := \mathbb{E}|X - X'|^3, \quad r := \frac{m}{2\rho}.$$

For $t \geq 0$, introduce the stop-loss functions

$$A(t) := \mathbb{E}(X - t)_+, \quad B(t) := \mathbb{E}(-X - t)_+,$$

and put $r_0 := \mathbb{E}X_+ = \mathbb{E}X_-$.

Lemma 2. *One has*

$$m = 2\rho + 12 \int_0^\infty (A(t)^2 + B(t)^2) dt, \tag{2.1}$$

$$3 \int_0^\infty A(t)^2 dt \leq r_0 \mathbb{E}X_+^2, \quad 3 \int_0^\infty B(t)^2 dt \leq r_0 \mathbb{E}X_-^2. \tag{2.2}$$

Consequently,

$$1 \leq r \leq 1 + \frac{\mathbb{E}|X|}{\rho} \leq 1 + \frac{1}{\rho} \leq 2. \tag{2.3}$$

If $\mu := \mathbb{E}X^3$, then

$$(r - 1)^2 + \left(\frac{\mu}{\rho}\right)^2 \leq 1. \tag{2.4}$$

Proof. For real x, y ,

$$|x - y|^3 = 6 \int_{\mathbb{R}} [(x - t)_+(t - y)_+ + (y - t)_+(t - x)_+] dt.$$

Taking expectations, using independence, splitting the integral at zero, and applying $\mathbb{E}X = 0$ gives

$$m = 12 \int_0^\infty (A^2 + B^2 + tA + tB) dt.$$

Fubini's theorem gives $\int_0^\infty tA(t) dt = \mathbb{E}X_+^3/6$ and the analogous identity for B , proving (2.1).

If $0 \leq y \leq z$, direct integration gives

$$3 \int_0^\infty (y - t)_+(z - t)_+ dt = \frac{3}{2}y^2z - \frac{1}{2}y^3 \leq \frac{1}{2}(yz^2 + y^2z).$$

Averaging independent copies proves (2.2). Since $\mathbb{E}X_+^2 + \mathbb{E}X_-^2 = 1$, it follows that $m - 2\rho \leq 4r_0 = 2\mathbb{E}|X|$, and (2.3) follows from $\mathbb{E}|X| \leq 1$ and Lyapunov's inequality $\rho \geq 1$.

To prove (2.4), write

$$a_j := \mathbb{E}X_+^j, \quad b_j := \mathbb{E}X_-^j \quad (j = 2, 3).$$

Positive semidefiniteness of the covariance matrix of (X_+, X_-) , together with $X_+X_- = 0$, gives $a_2b_2 \geq r_0^2$. Cauchy's inequality gives $a_2^2 \leq r_0a_3$ and $b_2^2 \leq r_0b_3$, hence $r_0 \leq \sqrt{a_3b_3}$. Therefore

$$m - 2\rho \leq 4\sqrt{a_3b_3} = 2\sqrt{\rho^2 - \mu^2}.$$

Division by 2ρ proves (2.4). □

The range in (2.3) will control the modulus of the characteristic function. Inequality (2.4) shows at the same time that a larger value of $r - 1$ forces the normalized third moment to be smaller. The next result transfers this tradeoff to the sine remainder at every frequency.

2.2 A bound for the sine remainder

The same parameter r controls $\mathbb{E}[\sin(uX) - uX]$ at every frequency.

Lemma 3 (Bound for the sine remainder). *For every $u \in \mathbb{R}$,*

$$|\mathbb{E}(\sin(uX) - uX)| \leq \frac{\rho|u|^3}{6} \sqrt{1 - (r - 1)^2}. \quad (2.5)$$

Proof. It is enough to take $u > 0$. Define

$$J := \int_0^\infty t(A + B) dt = \frac{\rho}{6}, \quad E_0 := \int_0^\infty (A^2 + B^2) dt, \quad H_u := \frac{1}{u} \int_0^\infty \sin(ut)(A - B) dt.$$

The point of this normalization is geometric. For a centered two-point law, the normalized quantities satisfy the unit-disk inequality $E_0^2 + H_u^2 \leq J^2$. Stop-loss functions are affine under mixtures, so Minkowski's inequality carries this constraint from two-point laws to an arbitrary centered law. We now make that mixture explicit.

Let α_+ be the restriction of the law of X to $(0, \infty)$, and let α_- be the reflection of its restriction to $(-\infty, 0)$. Centering gives

$$\int a \alpha_+(da) = \int b \alpha_-(db) = r_0,$$

and the variance-one assumption implies $r_0 > 0$. On $(0, \infty)^2$, define

$$\nu(da, db) := \frac{a + b}{r_0} \alpha_+(da) \alpha_-(db),$$

and add mass $\mathbb{P}\{X = 0\}$ at a zero component. For each (a, b) , let $X_{a,b}$ take the values a and $-b$ with probabilities $b/(a+b)$ and $a/(a+b)$. A direct calculation shows that the law of X is the ν -mixture of these centered two-point laws and the zero law. In particular, their stop-loss functions mix linearly to (A, B) .

For one such component, let $J_{a,b}, E_{a,b}, H_{a,b}$ denote the three quantities above, and put $x = ua, y = ub$. Direct calculation gives

$$\frac{E_{a,b}}{J_{a,b}} = \frac{2xy}{x^2 + y^2}, \quad \frac{H_{a,b}}{J_{a,b}} = -\frac{6(y \sin x - x \sin y)}{xy(x^2 + y^2)}.$$

For $z > 0$,

$$\left| \frac{d}{d(z^2)} \frac{\sin z}{z} \right| = \frac{|z \cos z - \sin z|}{2z^3} \leq \frac{1}{6},$$

because

$$|z \cos z - \sin z| = \left| \int_0^z t \sin t \, dt \right| \leq \int_0^z t^2 \, dt = \frac{z^3}{3}.$$

It follows that $E_{a,b}^2 + H_{a,b}^2 \leq J_{a,b}^2$. Applying Jensen's inequality at each t , together with linearity of the mixture, gives

$$E_0 \leq \int E_{a,b} \nu(da, db), \quad H_u = \int H_{a,b} \nu(da, db), \quad J = \int J_{a,b} \nu(da, db).$$

Minkowski's inequality in $\mathbb{R}^2$ now proves

$$\sqrt{E_0^2 + H_u^2} \leq \int \sqrt{E_{a,b}^2 + H_{a,b}^2} \nu(da, db) \leq J.$$

Two integrations of the sine remainder yield $\mathbb{E}(uX - \sin(uX)) = u^3 H_u$. By (2.1), $E_0/J = r - 1$. Thus the displayed bound on $E_0^2 + H_u^2$ gives $|H_u| \leq (\rho/6)\sqrt{1 - (r-1)^2}$, proving (2.5). $\square$

Thus r controls the sine remainder uniformly in the frequency. Together with the modulus estimate below, this gives a joint constraint on the complex characteristic-function error.

2.3 Constraints involving the first absolute moment

Write

$$s := \mathbb{E}|X|, \quad d := 1 - s, \quad \eta := \rho(r - 1).$$

The Hardy bounds (2.2), before replacing s by 1, give

$$m - 2\rho \leq 4r_0(\mathbb{E}X_+^2 + \mathbb{E}X_-^2) = 2s, \quad \text{hence} \quad \eta \leq s.$$

Cauchy-Schwarz also gives

$$1 = (\mathbb{E}|X|^2)^2 \leq (\mathbb{E}|X|)(\mathbb{E}|X|^3) = s\rho, \quad s \leq 1.$$

Consequently,

$$0 \leq d \leq \bar{d} := \min(1 - \rho^{-1}, 1 - \eta). \quad (2.6)$$

The identity $\mathbb{E}(|X| - 1)^2 = 2d$ explains the role of this additional constraint: small d forces $|X|$ to be close to 1 in mean square. Expanding the characteristic function about $|X| = 1$ will therefore give a comparison with the symmetric two-point law.

3 Characteristic-function comparisons

Let $f(u) := \mathbb{E}e^{iuX}$. The smoothing inequality requires bounds on both $|f(u)|$ and $|f(u) - e^{-u^2/2}|$: the first controls the decay of the characteristic function of the sum, and the second its difference from the Gaussian characteristic function. Put

$$g(v) := \cos v - 1 + \frac{v^2}{2}, \quad \kappa := \max_{v>0} \frac{g(v)}{v^3}.$$

Exact Taylor enclosures and proved derivative-sign bounds give

$$0.09916191350 < \kappa < 0.09916191353$$

and a unique maximizer ϑ satisfying

$$3.99589567 < \vartheta < 3.99589570.$$

We use Prawitz's piecewise characteristic-function minorant [6], at the symmetrized scale $r\rho$:

$$q(v) := \begin{cases} \frac{1}{2} - \kappa v, & 0 \leq v \leq \vartheta, \\ \frac{1 - \cos v}{v^2}, & \vartheta < v \leq 2\pi, \\ 0, & v > 2\pi. \end{cases} \quad (3.1)$$

Lemma 4. *The function q is convex and nonincreasing, and*

$$q(v) \leq \frac{1 - \cos v}{v^2} \quad (v > 0).$$

Consequently,

$$|f(u)|^2 \leq 1 - 2u^2 q(r\rho|u|). \quad (3.2)$$

Proof. The definition of κ gives the minorant on the first interval. At ϑ , equality and equality of the first derivatives hold. For $\psi(v) = (1 - \cos v)/v^2$,

$$\psi''(v) = \frac{v^2 \cos v - 4v \sin v + 6(1 - \cos v)}{v^4}.$$

The numerator is nonnegative on $[\vartheta, 2\pi]$, and $\psi'(2\pi) = 0$. The exact sign checks are part of the certificate in Section 5. Hence the three pieces join with nondecreasing slopes and q is convex and nonincreasing.

Let $Y = X - X'$. Since $\mathbb{E}Y^2 = 2$, the measure $Y^2 d\mathbb{P}/2$ is a probability measure. Jensen's inequality gives

$$\begin{aligned} 1 - |f(u)|^2 &= \mathbb{E}(1 - \cos(uY)) \\ &\geq 2u^2 q\left(|u| \frac{\mathbb{E}|Y|^3}{\mathbb{E}Y^2}\right) = 2u^2 q(r\rho|u|), \end{aligned}$$

which is (3.2). □

Estimate (3.2) will control the powers of f in the smoothing argument. We next combine (2.5) with the Taylor remainder bounds to obtain a one-step Gaussian comparison by planar maximization, retaining the same parameter r .

The sine-remainder bound gives an imaginary half-width $\beta(r)$, the ordinary third-order remainder gives a disk of radius d_0 , and the Gaussian correction shifts the comparison point along the real axis by $\epsilon_\rho(c)$. Set

$$\beta(r) := \frac{1}{6} \sqrt{1 - (r - 1)^2}, \quad d_0 := \frac{1}{6},$$

and, for $c \geq 0$, define

$$\epsilon_\rho(c) := \begin{cases} \frac{\rho^2}{c^3} \left(e^{-c^2/(2\rho^2)} - 1 + \frac{c^2}{2\rho^2} \right), & c > 0, \\ 0, & c = 0. \end{cases}$$

Let

$$\Xi(r) := \{0, \kappa\} \cup \left(\left\{ \frac{r-1}{6} \right\} \cap [0, \kappa] \right)$$

and

$$D(\rho, r, c)^2 := \max_{x \in \Xi(r)} \left\{ (x - \epsilon_\rho(c))^2 + \min(\beta(r)^2, d_0^2 - x^2) \right\}. \quad (3.3)$$

Thus $D(\rho, r, c)$ is the greatest possible distance from the Gaussian comparison point to the feasible Taylor-remainder region. The maximum can be computed by checking the real parts in $\Xi(r)$.

Lemma 5. *For every $u \in \mathbb{R}$, with $c = \rho|u|$,*

$$|f(u) - e^{-u^2/2}| \leq \rho|u|^3 D(\rho, r, c). \quad (3.4)$$

Proof. At $u = 0$ both sides are zero. For $u \neq 0$, set

$$z := \frac{f(u) - (1 - u^2/2)}{\rho|u|^3},$$

the definition of κ , Lemma 3, and the third-order Taylor remainder give

$$0 \leq \Re z \leq \kappa, \quad |\Im z| \leq \beta(r), \quad |z| \leq d_0.$$

Moreover,

$$\frac{e^{-u^2/2} - (1 - u^2/2)}{\rho|u|^3} = \epsilon_\rho(c).$$

Thus the normalized left side of (3.4) is the distance from the nonnegative real point $\epsilon_\rho(c)$ to a point in the intersection of a vertical strip, a horizontal strip, and the disk of radius d_0 . For fixed real part x , the largest squared imaginary part is $\min(\beta(r)^2, d_0^2 - x^2)$. The squared distance can change form only at $x = (r-1)/6$, and its maximum on each piece is attained at an endpoint. These are exactly the points in $\Xi(r)$, proving (3.3) and (3.4). $\square$

3.1 Comparison with the symmetric two-point law

The first absolute moment gives a second comparison, with $\cos u$. The following bounds express its error directly in terms of d , using the mean-square identity at the end of Section 2.3.

Lemma 6. *For every $u \in \mathbb{R}$,*

$$\begin{aligned} |\Re f(u) - \cos u| &\leq d(|u \sin u| + u^2), \\ |\Im f(u)| &\leq \sqrt{2d} |\sin u - u \cos u| + du^2. \end{aligned} \quad (3.5)$$

Proof. Let $Y = |X|$, and let $A \in \{-1, 1\}$ be the sign of X , with $A = 1$ when $X = 0$. Then $X = AY$, $\mathbb{E}AY = 0$, and $\mathbb{E}(Y-1)^2 = 2d$. Hence

$$|\mathbb{E}A| = |\mathbb{E}[A(Y-1)]| \leq \sqrt{2d}.$$

Global Taylor remainder bounds give

$$\begin{aligned} \cos(uY) &= \cos u - u \sin u(Y-1) + R_c, \\ \sin(uY) &= \sin u + u \cos u(Y-1) + R_s, \quad |R_c|, |R_s| \leq \frac{1}{2}u^2(Y-1)^2. \end{aligned}$$

Take expectations, using $\mathbb{E}(Y - 1) = -d$ and $\mathbb{E}[A(Y - 1)] = -\mathbb{E}A$. This gives

$$\Re f(u) - \cos u = du \sin u + \mathbb{E}R_c, \quad \Im f(u) = \mathbb{E}A(\sin u - u \cos u) + \mathbb{E}(AR_s).$$

Both remainder expectations have absolute value at most du^2 . $\square$

Intersect these real and imaginary bounds with the constraints in the proof of Lemma 5. This bounds both $|f(u)|$ and $|f(u) - \cos u|$. The latter gives the additional estimate

$$|f(u)^n - e^{-nu^2/2}| \leq n|f(u) - \cos u| \max(|f(u)|, |\cos u|)^{n-1} + |\cos(u)^n - e^{-nu^2/2}|. \quad (3.6)$$

Thus comparison with the two-point law also bounds the error in the normal approximation.

When r is close to 2, the moment constraints force d to be small uniformly in ρ . The resulting exponential modulus bound will control the powers of f in the large-sample estimate.

Lemma 7. *If $r \geq 19/10$ and $0 \leq u \leq 5/6$, then*

$$|f(u)| \leq e^{-(39/100)u^2}. \quad (3.7)$$

Proof. From (2.6),

$$d \leq 1 - \max(\rho^{-1}, \rho(r - 1)) \leq 1 - \sqrt{r - 1}.$$

The last inequality follows by considering whether $\sqrt{r - 1} \leq \rho^{-1}$. Since $(237/250)^2 \leq 9/10$, we have $d \leq 13/250$. Put $x = u^2$ and

$$P(x) = 1 - \frac{99}{250}x + \left(\frac{1}{24} - \frac{13}{1500}\right)x^2 + \frac{13}{30000}x^3.$$

Taylor bounds for sine and cosine applied to (3.5) give

$$|\Re f(u)| \leq P(x), \quad |\Im f(u)| \leq \frac{71}{500}x.$$

For the first bound, $1 - u^2/2 > 0$ is a lower bound on $\Re f(u)$. For the second, use $0 \leq \sin u - u \cos u \leq u^3/3$, $\sqrt{2}d \leq 81/250$, and $u \leq 5/6$; indeed, $(81/250)(5/6)/3 + 13/250 = 71/500$.

Let $Q_3(x) = 1 - \frac{39}{50}x + \frac{1}{2}(\frac{39}{50}x)^2 - \frac{1}{6}(\frac{39}{50}x)^3$. For $0 \leq x \leq 25/36$,

$$Q_3(x) - P(x)^2 - (71x/500)^2 = x q_*(x),$$

where

$$q_*(x) = \frac{3}{250} + \frac{3061}{50000}x - \frac{40367}{750000}x^2 - \frac{3729}{5000000}x^3 - \frac{143}{5000000}x^4 - \frac{169}{900000000}x^5.$$

Here $q_*'' < 0$, $q_*(0) = 3/250 > 0$, and

$$q_*(25/36) = \frac{98569207799}{3482851737600} > 0.$$

Concavity proves $q_* \geq 0$ throughout the interval. The alternating exponential bound $Q_3(x) \leq e^{-(39/50)x}$, followed by a square root, proves (3.7). $\square$

3.2 The range of the Gaussian correction

For $r > 1$, admissibility restricts $1 \leq \rho \leq 1/(r-1)$. At fixed $c \geq 0$, $\epsilon_\rho(c)$ decreases with ρ . For $c > 0$, this follows by writing it as

$$\frac{e^{-y} - 1 + y}{2cy}, \quad y = \frac{c^2}{2\rho^2},$$

since the derivative of $(e^{-y} - 1 + y)/y$ is $[1 - (1 + y)e^{-y}]/y^2 \geq 0$. The case $c = 0$ is direct. Thus

$$\epsilon_{1/(r-1)}(c) \leq \epsilon_\rho(c) \leq \epsilon_1(c).$$

The squared distance in (3.3) is convex in the Gaussian correction. Maximizing over $\Xi(r)$ preserves convexity, so the two feasible endpoints give

$$D(\rho, r, c) \leq D_*(r, c) := \max\{D(1/(r-1), r, c), D(1, r, c)\}. \quad (3.8)$$

This is used only when $r \geq 19/10$. Otherwise we use $D_0(r, c)$, the bound obtained from the larger interval $[0, \epsilon_1(c)]$ by evaluating the same squared distance at its endpoints.

4 Smoothing and the 0.4395 bound

Set $L := \rho/\sqrt{n}$. When L is large, no characteristic-function estimate is needed. The following bound uses only the first two moments.

Lemma 8. *If W is centered and has variance one, then*

$$\sup_x |\mathbb{P}\{W \leq x\} - \Phi(x)| \leq \frac{109}{200}.$$

Proof. At $x = 0$, the absolute error is at most $1/2$. For $a > 0$, Cantelli's inequality gives

$$\mathbb{P}\{W \geq a\} \leq \frac{1}{1+a^2}, \quad \mathbb{P}\{W \leq -a\} \leq \frac{1}{1+a^2}.$$

On each half-line one signed CDF error is at most $1/2$. For the other, use the displayed inequalities and, on the positive half-line, $\mathbb{P}\{W > a\} \leq \mathbb{P}\{W \geq a\}$. By symmetry of Φ , it suffices to bound $1/(1+a^2) - \Phi(-a)$. For $a \geq 1$ this is at most $1/2$. For $0 < a \leq 1$, the bound $(2\pi)^{-1/2} < 2/5$ gives

$$\frac{1}{1+a^2} - \Phi(-a) \leq \frac{1}{1+a^2} - \frac{1}{2} + \frac{2a}{5} \leq \frac{109}{200}.$$

The last inequality is equivalent to $p(a) := 9 - 80a + 209a^2 - 80a^3 \geq 0$. The following exact decompositions prove it on the two indicated intervals:

$$\begin{aligned} p(a) &= 20a^2(1-4a) + \frac{(189a-40)^2 + 101}{189}, & 0 \leq a \leq \frac{1}{4}, \\ p(a) &= \frac{13}{16} + (a - \frac{1}{4})[(a - \frac{1}{4})(169 - 80a) + \frac{19}{2}], & \frac{1}{4} \leq a \leq 1. \end{aligned}$$

Every term on the right is nonnegative on its stated interval. □

Put $C = 879/2000$ and $R = 1090/879$. Since

$$CR = \frac{109}{200}, \quad \frac{56}{45} - R = \frac{58}{13185} > 0,$$

Lemma 8 proves Theorem 1 when $L \geq R$. We certify the slightly larger complementary domain $L \leq 56/45$ using the characteristic-function bounds and Prawitz smoothing.

For $0 < t < 1$, define the Prawitz kernel

$$\begin{aligned} K(t) &:= \frac{1-t}{2} + \frac{i}{2} \left((1-t) \cot(\pi t) + \frac{1}{\pi} \right), \\ K(-t) &:= \overline{K(t)}. \end{aligned} \tag{4.1}$$

The Prawitz smoothing inequality converts characteristic-function estimates into a bound on Kolmogorov distance [5]. The normalization below is equivalent to the filter normalization recorded in [4, Section 2].

Lemma 9 (Prawitz smoothing inequality). *Let Y be an integrable real random variable, with distribution function F and characteristic function h . For $T > 0$ and $0 < t_0 < 1$,*

$$\begin{aligned} \sup_x |F(x) - \Phi(x)| &\leq 2 \int_0^{t_0} |K(t)| |h(Tt) - e^{-T^2 t^2/2}| dt \\ &\quad + 2 \int_{t_0}^1 |K(t)| |h(Tt)| dt \\ &\quad + 2 \int_0^{t_0} \left| K(t) - \frac{i}{2\pi t} \right| e^{-T^2 t^2/2} dt \\ &\quad + \frac{1}{\pi} \int_{t_0}^\infty e^{-T^2 t^2/2} \frac{dt}{t}. \end{aligned} \tag{4.2}$$

Proof. Let $J(x) = (\sin(\pi x)/(\pi x))^2$, with its removable value at zero, and let H be the odd band-limited function in Prawitz's construction. The pointwise estimate $|\operatorname{sgn}(x) - H(x)| \leq J(x)$ gives

$$B_-(x) := \frac{1 - H(x) - J(x)}{2} \leq \mathbf{1}_{(-\infty, 0]}(x) \leq \frac{1 - H(x) + J(x)}{2} =: B_+(x).$$

The nonconstant Fourier transforms of B_+ and B_- are supported on $[-1, 1]$; under the normalization in (4.1), they are $K(t)$ and $-K(-t)$, respectively.

Apply these bounds to $T(Y - x)/(2\pi)$, take expectations, and use Fourier inversion. The finite first moment removes the apparent singularity at zero:

$$|h(u) - 1| \leq |u| \mathbb{E}|Y|,$$

so the zero-frequency constant may be subtracted before passing to the symmetric Fourier limit. Subtracting the exact normal inversion formula, splitting the frequencies at t_0 , and using conjugate symmetry gives the first three terms in (4.2); the remaining normal frequencies give the last integral. Applying the same argument to $-Y$ supplies the opposite CDF inequality and completes the absolute-value bound. $\square$

We apply Lemma 9 to the law of S_n , which is integrable under the third-moment assumption. With $L = \rho/\sqrt{n}$ as above, choose

$$T := \frac{2\pi}{rL}, \quad t_0 := \frac{19}{50}. \tag{4.3}$$

4.1 The finite-sample envelope

For $t \geq 0$, write $u = 2\pi t/(r\rho)$, $B = e^{-u^2/2}$, and $N = N(t) := B^n = e^{-T^2 t^2/2}$. Set $\kappa_+ = 9916191353/10^{11}$. Unlike the exact κ in q and D , κ_+ is a certified rational upper bound, used in the real strip below.

First intersect the component bounds from Section 3. With $\bar{d}$ from (2.6), the interval $[a_u, b_u]$ contains $\Re f(u)$, and j_u bounds $|\Im f(u)|$, where

$$\begin{aligned} A_u &= \bar{d}(|u \sin u| + u^2), \\ a_u &= \max\{-1, \cos u - A_u, 1 - u^2/2\}, \\ b_u &= \min\{1, \cos u + A_u, 1 - u^2/2 + \kappa_+ \rho u^3\}, \\ j_u &= \min\{\sqrt{2\bar{d}}|\sin u - u \cos u| + \bar{d}u^2, u^3\sqrt{\rho^2 - \eta^2/6}\}. \end{aligned}$$

Combining these component bounds with the symmetrized modulus estimate gives $|f(u)| \leq F_u$, where

$$F_u = \min\{1, [1 - 2u^2 q(2\pi t)]_+^{1/2}, [\max(|a_u|, |b_u|)^2 + j_u^2]^{1/2}\}.$$

The same rectangle bounds the distance to $\cos u$: $|f(u) - \cos u| \leq E_u$, with

$$e_u = \min\{A_u, \max(|a_u - \cos u|, |b_u - \cos u|)\}, \quad E_u = (e_u^2 + j_u^2)^{1/2}.$$

The Gaussian comparison, the triangle inequality and the two-point comparison now give three bounds on $|f(u)^n - N(t)|$. Taking their minimum, define $M(t) = F_u^n$ and

$$\begin{aligned} R(t) &= \min\{n\rho u^3 D(\rho, r, 2\pi t/r) \max(F_u, B)^{n-1}, \\ &\quad F_u^n + N, \quad nE_u \max(F_u, |\cos u|)^{n-1} + |\cos(u)^n - N|\}. \end{aligned}$$

These bounds remain valid at $n = 1$ and at $t = 0$. Therefore $\Delta_n \leq \mathcal{U}_+(n, \rho, r)$, where

$$\begin{aligned} \mathcal{U}_+ &:= 2 \int_0^{t_0} |K(t)| R(t) dt + 2 \int_{t_0}^1 |K(t)| M(t) dt \\ &\quad + 2 \int_0^{t_0} |K_c(t)| N(t) dt + \frac{1}{\pi} \int_{t_0}^\infty N(t) \frac{dt}{t}, \end{aligned} \tag{4.4}$$

with $K_c(t) = K(t) - i/(2\pi t)$. Thus $\mathcal{U}_+$ bounds the smoothing functional using only n, ρ, r .

4.2 Large-sample reduction

For $n \geq 100$, define

$$Q(t) = \frac{(2\pi t)^2 q(2\pi t)}{r^2 L^2}, \quad Q_{\text{rate}}(t) = \frac{39(2\pi t)^2}{100r^2 L^2}, \quad \alpha(L) = \max\{99/100, 1 - L^2\}.$$

The two inequalities $n(99/100) \leq n - 1$ and $n(1 - L^2) = n - \rho^2 \leq n - 1$ prove

$$n\alpha(L) \leq n - 1. \tag{4.5}$$

Since the one-factor envelopes are nonnegative, an estimate $\max(F_u, B) \leq e^{-q_0}$, $q_0 \geq 0$, implies

$$\max(F_u, B)^{n-1} \leq e^{-\alpha(L)nq_0}.$$

On $r \geq 19/10$ and $t \leq 1/4$, one has $u \leq 5/6$, so Lemma 7 applies. Combining it with Lemma 4, put

$$Q_s(t) = \begin{cases} \max\{Q(t), Q_{\text{rate}}(t)\}, & t \leq 1/4, \\ Q(t), & t > 1/4. \end{cases}$$

The normalized low-frequency difference integrand is then at most

$$\frac{2|K(t)|(2\pi t)^3}{r^3 L^3} D_*(r, 2\pi t/r) e^{-\alpha(L)Q_s(t)}. \tag{4.6}$$

The estimate $u \leq 5/6$ uses only $\rho \geq 1$, $r \geq 19/10$ and the certified upper enclosure of π .

For $1/16 < L \leq 56/45$, the certificate uses the weaker fixed coefficient $99/100$ in (4.6). It takes the minimum of this bound and the Gaussian and triangle bounds obtained using D_0 and (3.2):

$$\frac{2|K(t)|}{L} \min \left\{ \frac{(2\pi t)^3}{r^3 L^2} D_0(r, 2\pi t/r) e^{-99Q(t)/100}, e^{-Q(t)} + N(t) \right\}.$$

The high-frequency and correction integrands are bounded, respectively, by

$$\frac{2|K(t)|}{L} e^{-Q(t)}, \quad \frac{2|K_c(t)|}{L} N(t).$$

A parameter interval not contained in $r \geq 19/10$ uses the bound with D_0 throughout.

For $0 < L \leq 1/16$, the variable coefficient in (4.6) is used when $r \geq 19/10$. The substitutions $t = Ly$ near zero and $t = 1 - Ly$ near one remove the apparent L^{-1} singularities before interval evaluation. Section 5 specifies the endpoint truncation and the exact allowance for the omitted intervals.

Let $\mathcal{P}_n$ denote the original right side of (4.2), with h the characteristic function of S_n and parameters (4.3). Then $\Delta_n \leq \mathcal{P}_n \leq \mathcal{U}_+$. The bounds above reduce the theorem to the following scalar inequality.

Lemma 10 (Exact continuous-domain bound). *For every admissible law and integer $n \geq 1$ such that*

$$1 \leq \rho \leq \frac{56}{45} \sqrt{n}, \quad 1 \leq r \leq 1 + \frac{1}{\rho}, \quad (4.7)$$

one has

$$\frac{\sqrt{n}}{\rho} \mathcal{P}_n < \frac{879}{2000}. \quad (4.8)$$

The finite certificates use (4.4); the large-sample certificates use the upper integrands just derived. The certificate is over the entire continuous parameter domain, not a sample of admissible laws.

Proof of Theorem 1. The standardized sum has mean zero and variance one by independence. If $L \geq 1090/879$, Lemma 8 gives

$$\Delta_n \leq 109/200 \leq (879/2000)L.$$

Otherwise $L < 1090/879 < 56/45$. The moment constraints place the law in (4.7). Lemmas 9 and 10 give $\Delta_n \leq \mathcal{P}_n < (879/2000)L$, which implies the stated weak bound. $\square$

5 Exact verification of the scalar bound

The preceding sections reduce the proof to Lemma 10. Its certificate proves an inequality over a continuum of parameters: finite data describe a covering, and a proved-sound interval checker verifies the desired bound on every member of that covering.

5.1 Exact interval bounds

A precision-48 dyadic interval $[l, h]$ represents $[l/2^{48}, h/2^{48}]$, with integer l, h . Rational conversion uses floor and ceiling; products use the outward four-corner hull. The verified operations also include division under a denominator sign condition, nonnegative square root and integer powers. Each operation has a containment theorem over the entire input interval.

For e^{-x} , $x \geq 0$, the checker halves the interval until its upper endpoint is at most $1/8$, uses degree-13 and degree-12 alternating Taylor bounds, and reverses the reduction by outward squaring. The Gaussian correction has a separate degree-25/24 enclosure. Sine and cosine on a full cell are enclosed using degree-33/32 midpoint polynomials, Lagrange remainder bounds, and the global Lipschitz bounds for their variation from the midpoint. Certified kernel bounds use the cotangent-gap expansion. The constants π, κ, ϑ enter through proved rational enclosures, not machine floating-point values.

For a partition $\mathcal{P}$ of J into closed intervals, let $[G](I)$ be an interval containing $G(t)$ for every $t \in I$. Then

$$\int_J G(t) dt \leq \sum_{I \in \mathcal{P}} |I| \sup[G](I).$$

Lean proves this implication with the actual outward-rounded multiplication and summation order of the checker. The normalized upper endpoint must be strictly below $879/2000$, and every interval-order, sign and analytic side condition must pass. Bisection uses two closed halves whose union is the parent, so shared endpoints cannot be omitted.

The improper normal tail is also enclosed. If $E_1(x) = \int_x^\infty e^{-v} dv/v$, then

$$\frac{1}{\pi} \int_{t_0}^\infty e^{-T^2 t^2/2} \frac{dt}{t} = \frac{E_1(T^2 t_0^2/2)}{2\pi}.$$

Using $E_1(x) = e^{-x} \int_0^\infty e^{-s} ds/(x+s)$, the checker applies the convex composite trapezoid bound on $[0, 24]$ with 768 intervals, then bounds the remaining integral by $e^{-24}/(x+24)$. Monotonicity replaces a positive interval parameter x by its lower endpoint in the upper direction.

5.2 Exhaustive regimes and endpoint limits

For each $1 \leq n < 100$, use

$$\rho = 1 + y \left(\frac{56}{45} \sqrt{n} - 1 \right), \quad r = 1 + \frac{z}{\rho}, \quad (y, z) \in [0, 1]^2.$$

Finite prefix codes describe complete binary subdivisions of this square. Each accepted leaf supplies a strict interval upper bound on its whole parameter cell. The union of the closed child cells covers each parent; the certificate is accepted only when this coverage and every leaf bound hold.

For $n \geq 100$, three closed parameter maps cover the regimes

$$0 < L \leq \frac{1}{16}, \quad \frac{1}{16} < L \leq \frac{1}{10}, \quad \frac{1}{10} < L \leq \frac{56}{45}.$$

In the first two maps the second coordinate is $r - 1 \in [0, 1]$. In the upper map it is $10\eta/\sqrt{n} = 10L(r - 1) \in [0, 1]$, and $r = 1 + z/(10L)$. The first coordinate affinely rescales L . The parameter interval is intersected with the proved constraint $r \leq 2$; this intersection preserves every admissible point.

For the small region, the endpoint substitutions retain $0 \leq y \leq 4$. The low, middle, high-endpoint and Gaussian-tail omissions have normalized upper bounds

$$\frac{13}{10^{12}}, \quad \frac{1}{10^{12}}, \quad \frac{1}{10^{12}}, \quad \frac{1}{10^{12}},$$

respectively. Their sum is below the allowance $3/10^{11}$. These are analytic bounds, proved using $e^{-32} \leq 1/(7 \cdot 10^{13})$, not unverified truncated tails. After the substitutions the endpoint expressions have certified extensions at $L = 0$. This covers arbitrarily small positive L without imposing an upper limit on n .

Two small-region trees split $r - 1$ at the exact dyadic $253327479039591/2^{48}$, which lies just above $9/10$. Their closed intervals meet at that point; equality is assigned to the lower tree. The equalities $L = 1/16$, $L = 1/10$ and $L = 56/45$ are included in the corresponding closed certificates. Thus the finite and large-sample cases exhaust (4.7).

5.3 Formal verification and reproducibility

The moment identities, characteristic-function bounds, smoothing theorem, interval containment and continuous coverage are formalized in Lean 4 [2] with Mathlib [8]. The accompanying source repository contains the complete theorem, exact certificate data and reproduction instructions. Its formalization guide maps the mathematical arguments to the Lean development.

The analytic implications and checker soundness are checked by the Lean kernel. Concrete Boolean evaluation uses `native_decide`, which additionally trusts Lean’s native compiler. There are no unproved placeholders or user-declared axioms. Only the finite certificate data, not the program that searches for them, enter the proof.

This verifies Lemma 10 on the full continuous domain. The mathematical improvement comes from retaining the first absolute moment in the characteristic-function comparisons and sharpening their use in the smoothing inequality. The certificates establish that these inequalities attain the stated universal constant.

AI disclosure

The proof, Lean formalization, exact numerical certificates, and initial manuscript were developed autonomously using OpenAI Codex, primarily with GPT-5.6 Sol. The authors supplied the research question and candidate materials and coordinated the review and preparation of the note.

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